TAHA AZIZ

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EXPERIENCE

Modelmouse Toronto, ON
Founding Software Engineer Jan. 2025 – Present

- Architected a production RAG system with hybrid retrieval and LLM-based inference, serving AI user personas at scale on AWS with FastAPI, PostgreSQL, and Redis
- Engineered horizontally scalable infrastructure across Docker deployments, successfully load-testing 1500+ concurrent connections with Locust, and achieving P99 latency under 350ms
- Built a data pipeline ingesting 10M+ behavioral event streams from Snowflake, applying chunking, vector embedding, semantic clustering, and schema validation to ensure AI model input quality
- Designed an eval harness measuring AI prediction accuracy at 79%, using precision and recall metrics and an LLM-as-judge based on human-labeled data
- Owned technical relationships with clients by leading demos, API integration, and monitoring production

Sentia Toronto, ON
Software Engineer Jul. 2024 – Dec. 2024

- Maintained Sentia data storage platform on Azure and Node.js for 50+ global enterprise clients
- Redesigned backup pipeline using chunk-based SHA-256 deduplication prior to Azure Blob Storage writes, lowering storage volume by 23% and saving ~$35k/year in cloud costs
- Managed Kubernetes clusters in production, including diagnosing pod logs to identify resource exhaustion and scaling node pools to restore service availability
- Engineered a Go microservice to handle 15,000+ metadata writes per second to PostgreSQL

Software Engineer Co-op Jan. 2023 – Aug. 2023

- Developed an LLM support assistant using Node.js and OpenAI, reducing support ticket volume by 10%
- Integrated OpenTelemetry tracing and metrics, reducing engineering time spent diagnosing crashes

Sheertex Montreal, QC
Software Developer Co-op May 2022 – Jul. 2022

- Engineered a QA pipeline with machine vision using Python and PyTorch, enabling automated defect detection at 96.7% accuracy for manufactured goods to reduce QA time by 75%

PROJECTS

Vis-jit NPM Package | JavaScript 2026

- Authored npm module “vis-jit” with JavaScript, for tracing and visualizing Node.js JIT optimizations, such as type inference, inline caching, and compiler IRs, currently serving 400+ weekly active downloads

Waterloo Rocketry: Rocket Tracking Software | C++ 2024

- Developed embedded C++ software for GPS radio telemetry, with checksum validation of incoming bytes for fault tolerance, allowing operation with up to 50% malformed byte frames

SKILLS

Languages: JavaScript, TypeScript, Python, SQL, C, C++, Go
Frameworks/Libraries: Node.js, FastAPI, PyTorch, pytest, pandas, NumPy, LangChain, OpenTelemetry
Tools/Infrastructure: Nginx, Linux, Docker, Kubernetes, Azure, AWS, Git, CI/CD, Claude Code
Databases: SQLite, PostgreSQL, Redis, Snowflake, Pinecone

EDUCATION

University of Waterloo Waterloo, ON
BASc in Mechatronics Engineering and Computing (GPA: 3.7/4.0) Jun. 2024

- Coursework: Operating Systems, Algorithms, GPU Programming, Pattern Recognition, Intro to AI